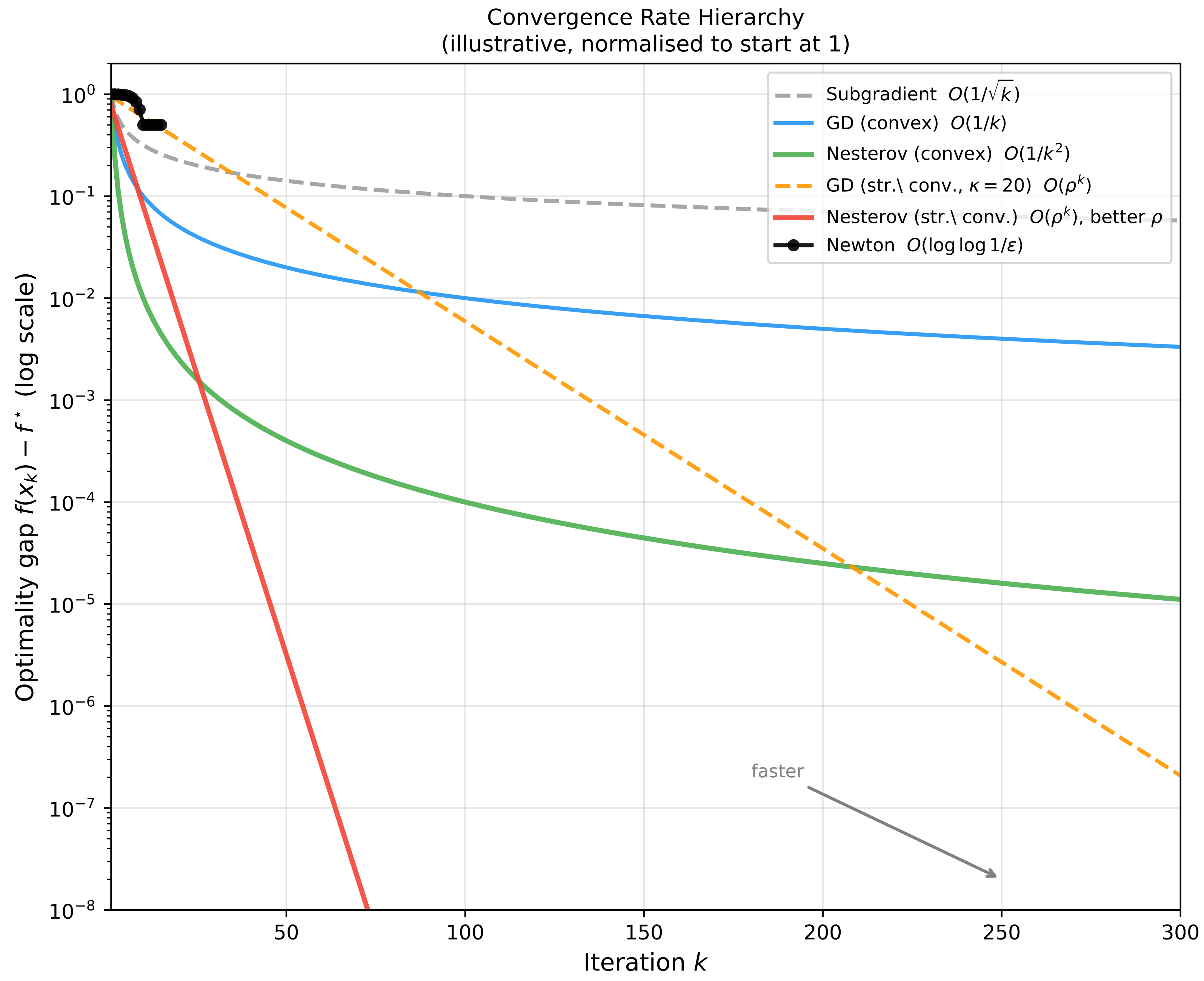

Algorithm Taxonomy: Rate, Cost, and Setting

Method	Setting	Rate	Cost/iter	Notes
Subgradient	Non-smooth	$O(1/\sqrt{k})$	$O(d)$	Optimal non-smooth
GD (convex)	Smooth	$O(1/k)$	$O(d)$	—
Proj. GD	Smooth+constrained	$O(1/k)$	$O(d)$ + proj	—
ISTA	Composite	$O(1/k)$	$O(d)$ + prox	$f + \psi$
Frank–Wolfe	Smooth+constrained	$O(1/k)$	$O(d)$ + LMO	Projection-free
Nesterov	Smooth	$O(1/k^2)$	$O(d)$	Optimal smooth
FISTA	Composite	$O(1/k^2)$	$O(d)$ + prox	Accelerated ISTA
GD (str. conv.)	Smooth+s.c.	$O(\kappa \log 1/\varepsilon)$	$O(d)$	Linear
Nesterov (s.c.)	Smooth+s.c.	$O(\sqrt{\kappa} \log 1/\varepsilon)$	$O(d)$	Optimal s.c.
SVRG/SAGA	Finite-sum+s.c.	$O((n+\kappa)\log 1/\varepsilon)$	$O(b)$	VR linear
Newton	Smooth+s.c.	$O(\log\log 1/\varepsilon)$	$O(d^3)$	Super-linear
L-BFGS	Smooth+s.c.	super-linear (heuristic)	$O(md)$	Quasi-Newton